# Towards Foundation Models for 3D PDEs

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https://ge.in.tum.de/

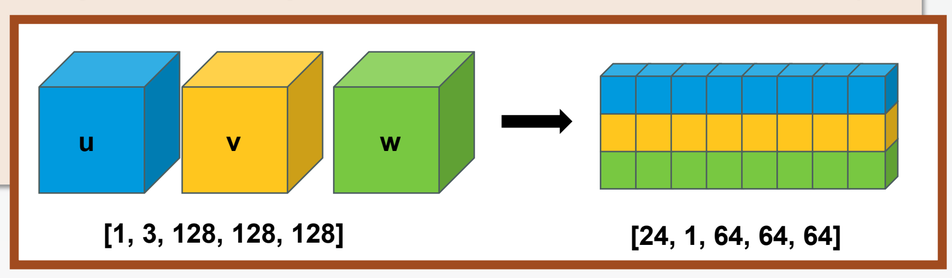
**We aim to develop foundation models for partial differential equations (PDEs): scalable neural networks pre-trained on diverse PDE datasets that can be efficiently adapted for new PDE problems and applications with higher accuracy and reduced training cost.**

## Regular Grids Model: Tadpole[1]

### Self-supervised pre-training

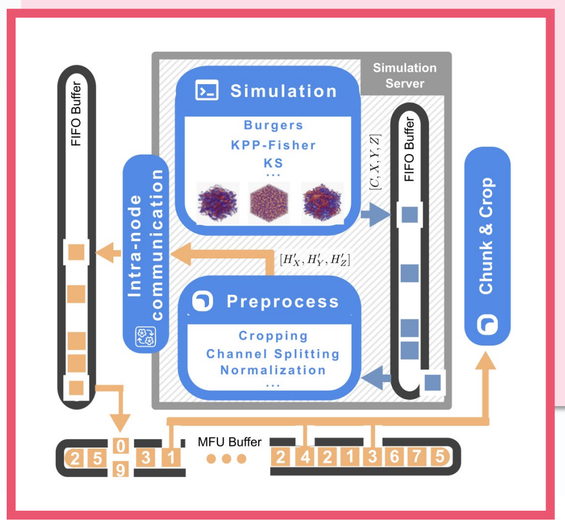
Instead of training on PDE dynamics, Tadpole is pre-trained with a **reconstruction objective**. This reconstruction pre-training
- eliminates the need for physical parameter embeddings.
- exhibits improved generalizability.
- supports a wide range of downstream tasks.

Tadpole is pre-trained on **single-channel crops** of simulation data, enabling both training and inference on data with arbitrary shapes.




### Online Data Generation

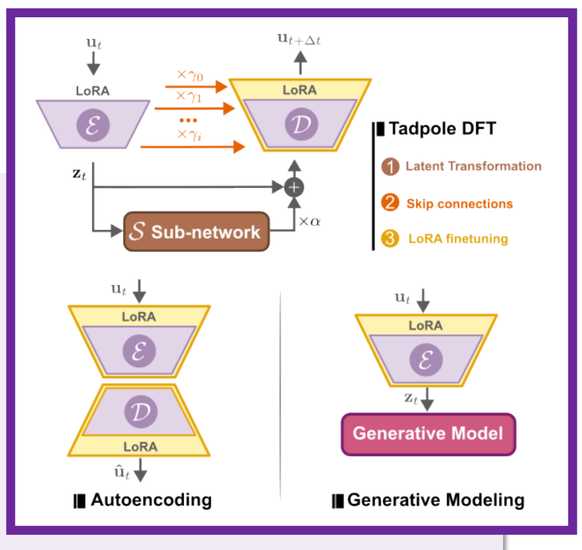
Tadpole is pre-trained using an **online learning** approach. GPU-based simulators run **in parallel** with the training process, supplying data through a carefully designed **buffer** system. without local storage.



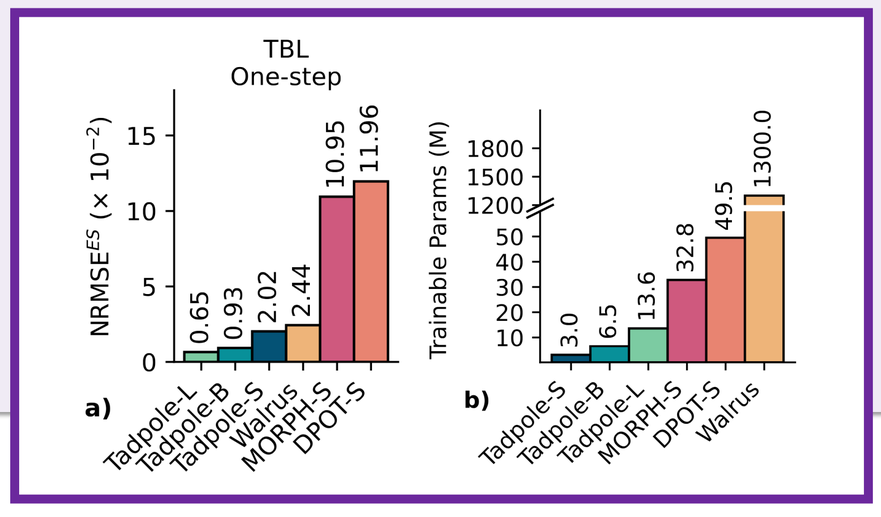

This buffer system decouples training from data simulation, preventing simulation from impacting training speed. The overall online-training framework provides a **10X** increase in training with over **200TB** of data

### Efficient fine-tuning

The pre-trained Tadpole can be adapted for **various** downstream tasks, such as autoencoding, generative modeling, and dynamics learning through **parameter-efficient fine-tuning**.




In particular, a Tadpole-DFT strategy is introduced for dynamics learning, enabling the transformation of a pre-trained autoencoder into a dynamics predictor. Using Tadpole-DFT, a **73% improvement** is achieved over Walrus, the previous state-of-the-art, while utilizing only **1% trainable parameters** for Navier-Stokes simulations, which are not included during pre-training.




## Unstructured Grids model: GMP[2]

### Methodology

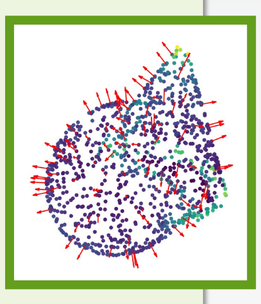
Same idea: Pre-training without disk-stored data, but…
…fast spectral solvers are not available in arbitrary domains

- **Online generation of primitive shapes**

Randomly generate intersections of cubes, spheres…

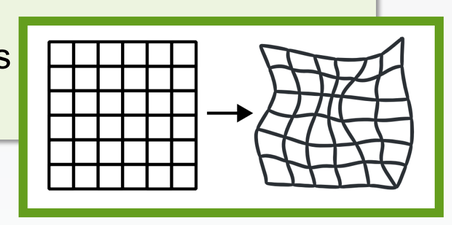
- **Geometry pre-training for steady simulations**

Take you model and pre-train it to predict geometric descriptors
  - Input: geometry coordinates
  - Output: normals, curvatures, VDF



- **Physics pre-training for autoregressive simulations**

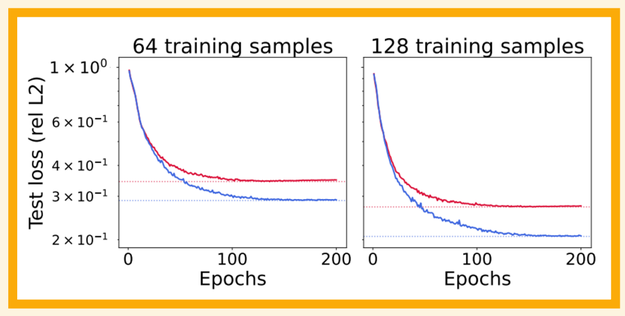
Similar to Tadpole: autoencode online solutions of PDEs interpolated in…
  - Surface: previous random geometries
  - Volume: Gaussian deformed grids

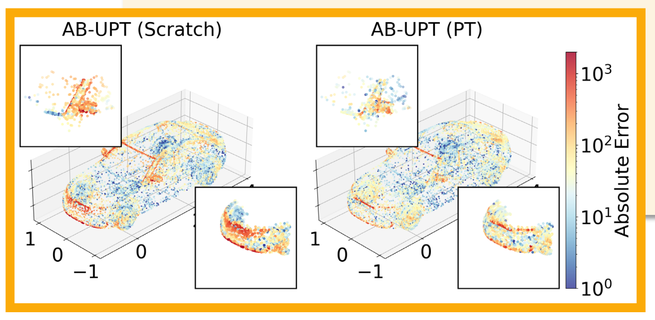


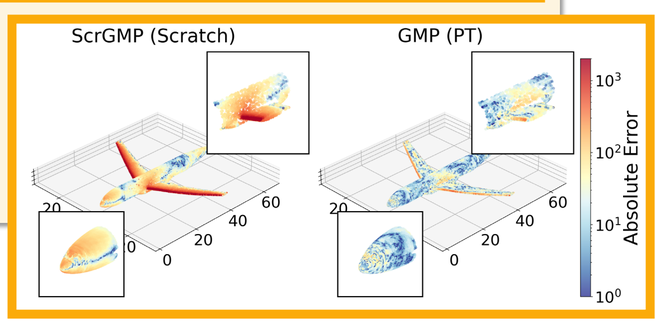
### Results: steady

Works well in several experiments
- Models: SMART, Transolver++, AB-UPT
- Datasets: DrivAerML, Shift-Wing, Shift-Crash

**Fine-tuning** with very **few training samples**
- Reduced error
- Faster convergence
- Data saving



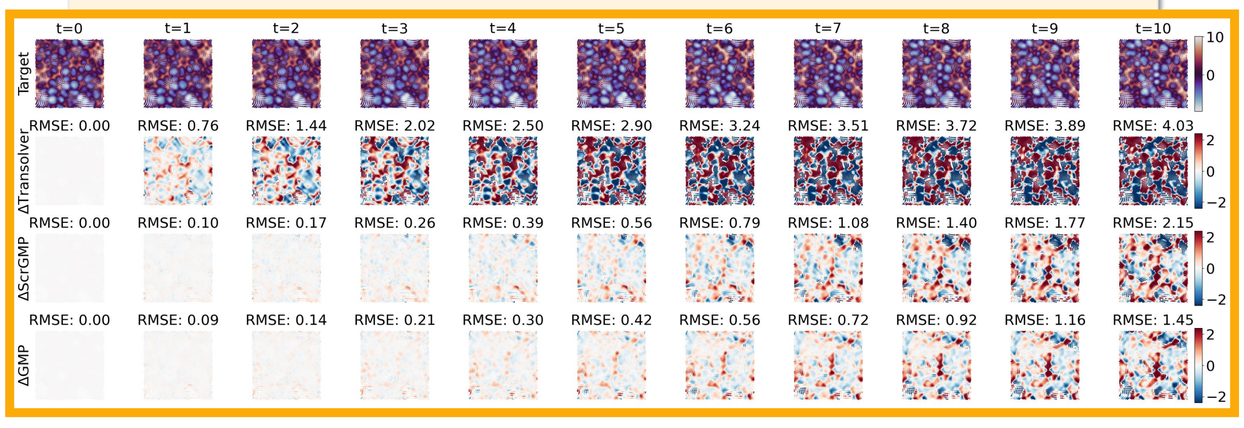




### Results: autoregressive

**Transient** simulations autoregressively with **top accuracy**

| | 2D KS ↓ | | | 2D Ellipse ↓ | | | 3D Shift-Crash ↓ | | |
|---|---|---|---|---|---|---|---|---|---|
| Timesteps | 1 | 5 | 10 | 1 | 5 | 20 | 1 | 5 | 10 |
| Erwin | 0.143 | 0.713 | 1.281 | 0.016 | 0.083 | 0.875 | 0.039 | 0.131 | 0.245 |
| Transolver++ | 0.203 | 0.779 | 1.028 | 0.006 | 0.021 | 0.063 | 0.020 | 0.072 | 0.072 |
| ScrGMP | 0.030 | 0.196 | 0.615 | 0.008 | 0.025 | 0.064 | 0.009 | 0.030 | 0.040 |
| GMP | 0.024 | 0.148 | 0.479 | 0.006 | 0.020 | 0.062 | 0.008 | 0.026 | 0.034 |
| Rel. Improvement | 20% | 24% | 22% | 25% | 20% | 3% | 11% | 13% | 15% |

NRMSE for different timesteps




[1] Tadpole: Autoencoders as Foundation Models for 3D PDEs with Online Learning. Q. Liu, F. Koehler, B. Holzschuh, N. Thuerey, arxiv:2605.15284

[2] Geometry Meets Physics: Data-Efficient Pre-Training for Unstructured Neural PDE Solvers. L. Medrano-Navarro, G. Baldan, Q. Liu, B. Holzschuh, J. Hagnberger, M. Niepert, N. Thuerey (coming soon!)

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